

# Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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## Abstract

Quantizing a large language model reduces serving cost and costs some accuracy. Determining how much normally requires re-running a full benchmark suite, which is the expense quantization was meant to avoid. We ask whether the accuracy delta can be predicted instead. Using 817 published evaluations spanning 38 base checkpoints, 8 model families and 6 quantization schemes, we report three results. First, a *negative* one: per-model prediction carries almost no signal. Under leave-one-family-out validation the best learned predictor beats a global mean by 0.03pp of MAE, and both ridge regression and gradient boosting are *worse* than guessing the average, because 33–100% of the variance on most benchmarks is evaluation noise. Second, a positive one: conditioning only on the quantization scheme yields split-conformal intervals with 90.1% empirical coverage (118/131, 95% CI [83.6, 94.6]) on checkpoints never used in construction. Third, and most consequential for practice: the published corpus is outcome-truncated. Every configuration in it is one that someone chose to release. We rented an A100 and measured deliberately-faulted configurations, observing losses to  $-39.5$ pp against a published worst case of  $-8.86$ pp—measured, we stress, only on models of 1.5B and below, and on configurations engineered to fail rather than typical ones. We argue the resulting artifact should be described as a calibrated historical baseline rather than a predictor, and we state precisely why its coverage guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and every figure’s provenance are public.

## 1 Introduction

Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8 wants one number: how much accuracy will this cost? The honest answer today is that nobody knows without running the evaluation.

This paper asks whether that number can be forecast from the public record. Model publishers release quantized checkpoints with benchmark tables comparing the compressed model against its dense parent. Aggregated, these are thousands of paired measurements of exactly the quantity of interest.

Our contributions:

1. **A negative result (§5).** Per-model prediction of the accuracy delta does not work, and we show why: the target is dominated by evaluation noise. We report this as the primary finding rather than burying it.
2. **A calibrated envelope that does work (§6),** conditioned on scheme alone, with measured coverage on held-out families and an explicit rule that withholds an interval where coverage is poor or the evidence is too thin to judge.

3. **A measurement of the corpus’s own selection bias** (§7), via GPU experiments on deliberately-faulted configurations—data the published record structurally cannot contain.
4. **An audit** (§9) recording every error we found in our own work, including two that changed published claims.

## 2 Related Work

**Predicting evaluations.** BenchPress [9] establishes the mechanism closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is a model’s absolute score; ours is the paired delta between a model and its compressed self, and their corpus contains no quantized checkpoints.

**Conformal prediction on compressed models.** Tong et al. [6] apply conformal prediction directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand differs: their conformal sets are over label space, constructed from a compressed model’s own output probabilities, so the method measures uncertainty *after* running the model. Ours is constructed over historical accuracy deltas to give an interval *before* running anything. Their Figure 2 plots the same quantity we predict,  $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$ , but measures it. Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can claim (§8).

**Quantization methods.** The checkpoints we harvest are produced with GPTQ [2] and SmoothQuant [8] via llm-compressor [7].

**The source of our data.** Kurtić et al. [5] report over 500,000 evaluations of quantized Llama-3.1 models, finding recovery above 99% of baseline. That work does not address whether the published set of quantized models is a biased sample. Characterising that gap is the contribution of §7.

**Selection bias.** Survivorship bias in benchmark corpora has been formalised and quantified in information retrieval [3]. Conformal prediction’s guarantee is known to fail under selection: conditional on a selection event, calibration data are no longer exchangeable with a test point [1, 4]. We apply this directly to our own artifact in §8.

## 3 Data

We harvest published model cards into rows of (model, scheme, benchmark,  $\text{acc}_{\text{before}}$ ,  $\text{acc}_{\text{after}}$ ), yielding 817 rows after filtering, across 38 base checkpoints, 8 families and 6 schemes.

**An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair. This is redundant information, and we exploit it: for every three-column row we check  $\text{recovery} \approx 100 \times \text{after}/\text{before}$  to a tolerance derived from the card’s own printed precision, and reject rows that disagree. This caught fabricated values arising from column-order variation between publishers, and internally inconsistent cards. Rejections are recorded rather than silently dropped.

**A structural limitation, stated at the outset.** All rows come from a single publisher using a single toolchain. Holding out a model family tests transfer across *models*; nothing in this design tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

## 4 Method

**Split conformal.** Given calibration residuals  $\{s_i\}_{i=1}^n$  and level  $\alpha$ , the interval half-width is the  $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which gives exact finite-sample coverage under exchangeability. Where  $n$  is too small for that index to exist, we return  $\infty$  rather than a misleadingly finite band.

**Mondrian (group-conditional) calibration.** A single marginal interval advertised at 90% was in fact a 68% interval for NVFP4 users—the users who most need it. We therefore calibrate per scheme.

**A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and adequately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is measurably poor on a sufficient number of scored rows. It is *insufficient evidence*, again with no interval, when the cell rests on fewer than 3 distinct checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on its coverage straddles the 85% line. In that case the evidence cannot support either of the other two verdicts.

Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound. The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one to anything, so counting that as a failure would flag cells that are merely outperforming. Comparing a one-sided measurement against the printed 90% is the *lenient* reading, not a flattering one: the interval is symmetric, so the permitted 10% is split across two tails, and pooled one-sided coverage across all scored rows is 95.1%.

Two cells were flagged under the earlier two-state rule, and neither is a confident refusal under the three-state one. `w4a16|<2B` had one-sided coverage of 70.1% over 77 rows (two-sided 68.8%), but those rows come from only 2 checkpoints. `w8a16|>10B` had 81.1% over 217 rows from 5 checkpoints (two-sided 73.7%), and its bootstrap interval,  $[47, 100]\%$ , straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those two cells; applying the rule to every cell moved 8 of 17 into that state, including several whose point-estimate coverage looked comfortable. The one-sided scoring and the three-state rule were proposed by an external commenter who identifies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed each of their figures from the raw rows before adopting either change, and one of their descriptions did not survive that check.

**A support floor.** Coverage measured on many benchmarks from one checkpoint is not independent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

## 5 The negative result

Under leave-one-family-out validation, row-weighted MAE:

| predictor                       | MAE (pp) | beats global mean? |
|---------------------------------|----------|--------------------|
| per-scheme mean (shipped)       | 0.7227   | yes                |
| global mean (baseline)          | 0.7545   | —                  |
| per-(scheme $\times$ benchmark) | 0.7607   | <b>no</b>          |
| ridge, 38 features              | 0.7639   | <b>no</b>          |
| per-benchmark mean              | 0.7647   | <b>no</b>          |
| gradient boosting               | 0.7878   | <b>no</b>          |

Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size, bit-width, method, benchmark identity and headroom all degraded out-of-family performance.

**Why.** Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta should be approximately zero—gives a noise floor that accounts for 33–100% of observed variance on most benchmarks. On GSM8K the noise alone is  $\pm 2\text{pp}$ , while the quantity being predicted has mean  $-0.36\text{pp}$ . The target is a difference of two noisy quantities that largely cancel: the numerically smallest and hardest-to-predict summary of quantization damage available.

**Implication.** The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band, with no model and no conformal machinery, achieves comparable coverage. What conformal contributes is a finite-sample guarantee and a principled rule for small- $n$  schemes, not accuracy. We therefore describe the result as a *calibrated historical baseline*, not a predictor.

## 6 Results

| scheme      | 90% interval (pp) | worst observed | % losing $>3\text{pp}$ | evals |
|-------------|-------------------|----------------|------------------------|-------|
| fp8         | $[-1.11, +0.48]$  | -1.90pp        | 0.0%                   | 80    |
| w8a8_int    | $[-1.71, +1.07]$  | -3.32pp        | 1.0%                   | 194   |
| fp8_dynamic | $[-1.44, +1.34]$  | -8.72pp        | 1.0%                   | 200   |
| w8a16       | $[-0.98, +0.81]$  | -3.04pp        | 1.2%                   | 86    |
| w4a16       | $[-2.87, +1.40]$  | -8.86pp        | 5.7%                   | 193   |
| nvfp4       | $[-4.10, +1.87]$  | -8.12pp        | 15.6%                  | 64    |

**Prospective validation.** After freezing the method we harvested checkpoints not used during development. Under a strict protocol that excludes any family present in training and any card whose format informed a parser fix, coverage is  $118/131 = 90.1\%$ , 95% Clopper–Pearson CI  $[83.6, 94.6]$ , against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code with the pipeline reproduces this figure with zero disagreements.

**What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when real damage occurs, the interval usually misses it. This is the single most important caveat in the paper.

## 7 The measured left tail

The corpus contains only configurations someone chose to publish. To measure what it structurally cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

**Result.** Worst measured loss  $-39.5\text{pp}$ , against  $-8.86\text{pp}$  as the worst loss anywhere in the published corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor instead of per-group.

**Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes that catastrophic damage is reachable; it is not a measured bound for larger models.

**A confound we found and did not resolve.** Re-running MMLU under the published protocol reproduces published base accuracy (47.20 against 47.42; 59.57 against 60.98), closing the harness question. It exposed a larger one: our own *correctly-configured* control is  $2.6\times$  more damaging than published W4A16 on closely comparable checkpoints (mean gap  $-3.78\text{pp}$ ,  $z = 2.9$  and  $5.3$ ). The cause is approximately  $171\times$  less calibration data than production defaults, not a bug. Consequently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It is a conservative anchor, not a bound, and we decline to publish a corrected point estimate: the correction moves by  $5.70\text{pp}$  depending on the anchor set, and we have only 6 rows from two models.

## 8 Limitations

**The guarantee is conditional, and void where it matters most.** Split conformal gives exact coverage under exchangeability between calibration and test. Our calibration set is the set of configurations a publisher chose to release—a selection event. Conditional on selection, calibration points are not exchangeable with an arbitrary test point and the guarantee does not transfer [1, 4]. It holds for a new checkpoint drawn from the same publication process. It is **void** for a recipe the reader tuned themselves and that nobody would have published—which is the case a practitioner most wants covered.

**Accuracy is not the whole of deployment risk.** Tong et al. [6] show compression frequently decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none of that.

**One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence that practitioner and calibration volume can dominate scheme choice. We measured how confining this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9 publishers carry paired before/after evaluations, so others do publish them. What the other publishers almost never provide is a *Recovery* column, which is what lets each row be checked against  $100\times$  after/before and rejected on disagreement: of their rows, 6 pass that gate, all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones published in a self-verifying form.

**Rows are not independent.** Up to sixteen benchmark rows come from a single quantization run. We mitigate with group-conditional calibration and the support floor, and note that a cluster bootstrap over checkpoints does not materially widen the headline interval, but the unit of analysis remains a row.

## 9 Audit

We record errors found in our own work, including those that changed published claims. This section exists because the alternative—reporting only what survived—would misrepresent how the result was obtained.

### Errors that changed a published number.

- *A retracted headline.* An early claim that held-out models came from families never used in construction was wrong; they were new checkpoints of families already present. The strict figure is 118/131.
- *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was matched as `Math-Lvl-5`, contaminating 7 test rows with scores from a different benchmark.
- *A centring mismatch.* Calibration measured residuals about the stratum mean while intervals centred on the scheme mean, mis-sizing every calibrated width. Found only by independent reimplementations.
- *A column-order fault.* One publisher orders columns [recovery, base, quant], producing a fabricated  $-68\text{pp}$  delta. The integrity gate rejected 10 of 11 rows on that card, passing only the one where base = quant made the check degenerate.

**A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band covered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch. On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-sided empirical index requires more calibration points than are often available. An infinite interval covers by construction. On rows where both are finite, the original wins on coverage *and* width. The change was reversed.

**An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without qualification. Checking the method against the conformal literature rather than against our own earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected in §8.

**Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in the test set. Rather than silently retain them, we record which fixes were test-informed and exclude the affected family from the strict headline.

**Mechanical verification.** Every figure in this paper is a macro generated from a claims registry that recomputes each value from source artifacts; no number here is hand-typed. The accompanying repository enforces this with a test that fails if any outward-facing document contains an untraced numeric literal.

## 10 Conclusion

Per-model prediction of quantization accuracy damage does not work on published data, and the reason is measurement noise rather than model capacity. What does work is narrower: a per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints, that declines to answer where its own measured coverage is poor or too thinly supported to judge. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured losses roughly four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

## References

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